



UNIVERSITY OF KHARTOUM • FACULTY OF ENGINEERING • ELECTRICAL & ELECTRONIC ENGINEERING

Improving Arabic RAG Retrieval Quality with LLM-based Query Enhancement

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AGENDA

01

Introduction



04

Methodology



02

**Problem &
Research Question**



05

**Results &
Discussion**



03

Objectives



06

**Conclusion &
Future Work**





INTRODUCTION • A LANGUAGE MODEL IS FROZEN IN TIME

Parametric memory

Frozen at training time

Retraining is expensive

Hallucination



Everything it knows is sealed inside

? A question it was never trained on

“ ما الشروط التي تسمح للطلاب بتأجيل دراسته في جامعة الخرطوم؟ ”

THE MODEL'S ANSWER

“ يجوز للطلاب تأجيل دراسته لمدة تصل إلى خمس سنوات بموجب المادة ١٢ من لائحة عام ١٩٩٨، بعد سداد رسوم قدرها ٥٠٠ جنية. ”

Invented. No such article, no such fee.

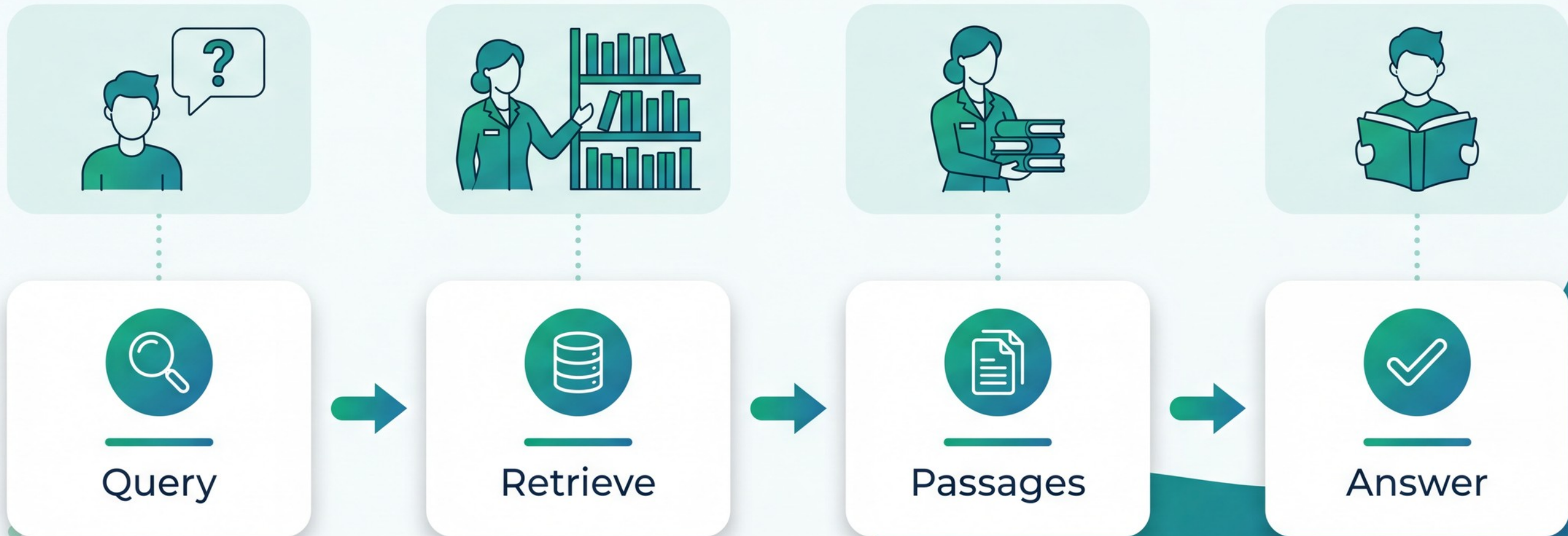


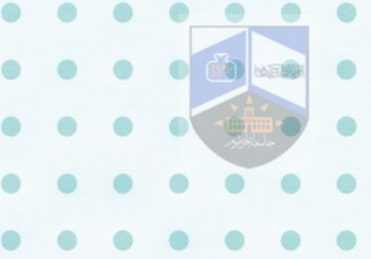
INTRODUCTION •

DON'T RETRAIN THE MODEL. GIVE IT THE DOCUMENTS.

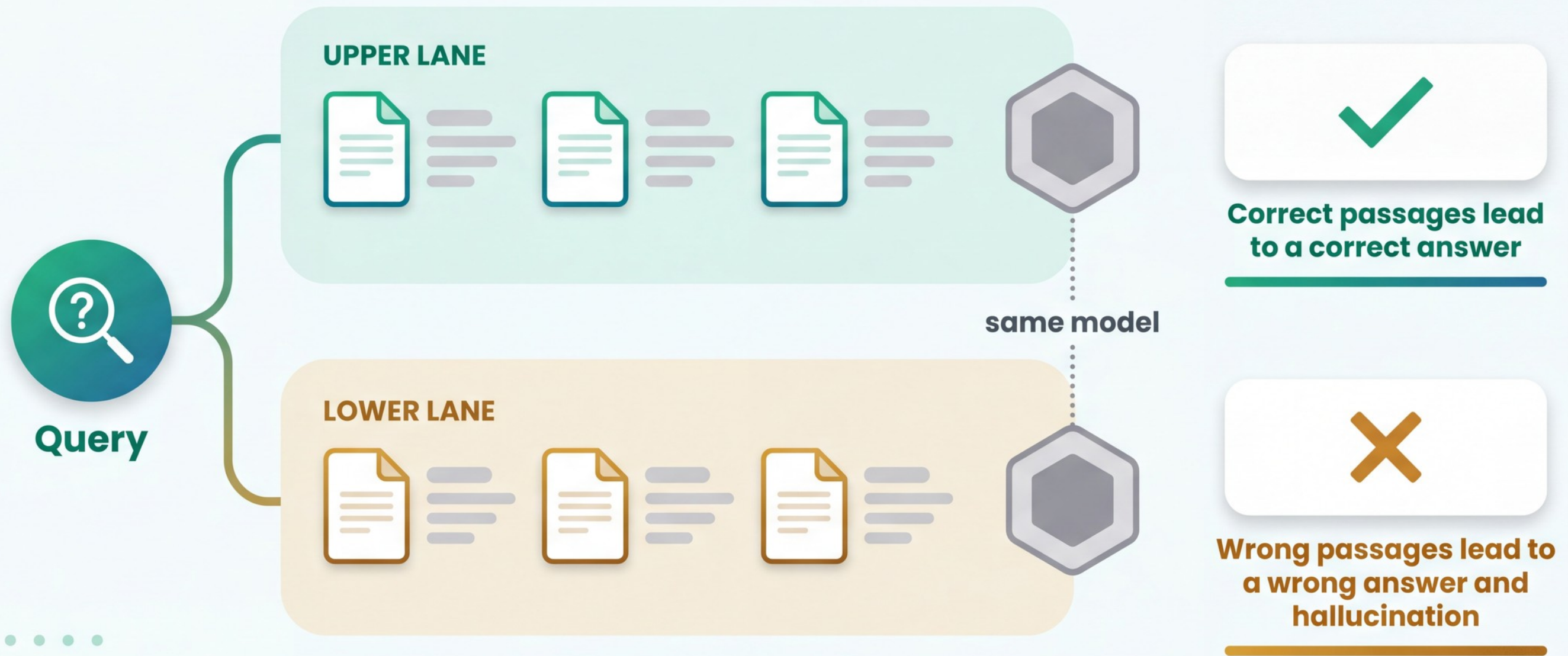
RAG

Retrieval-Augmented Generation





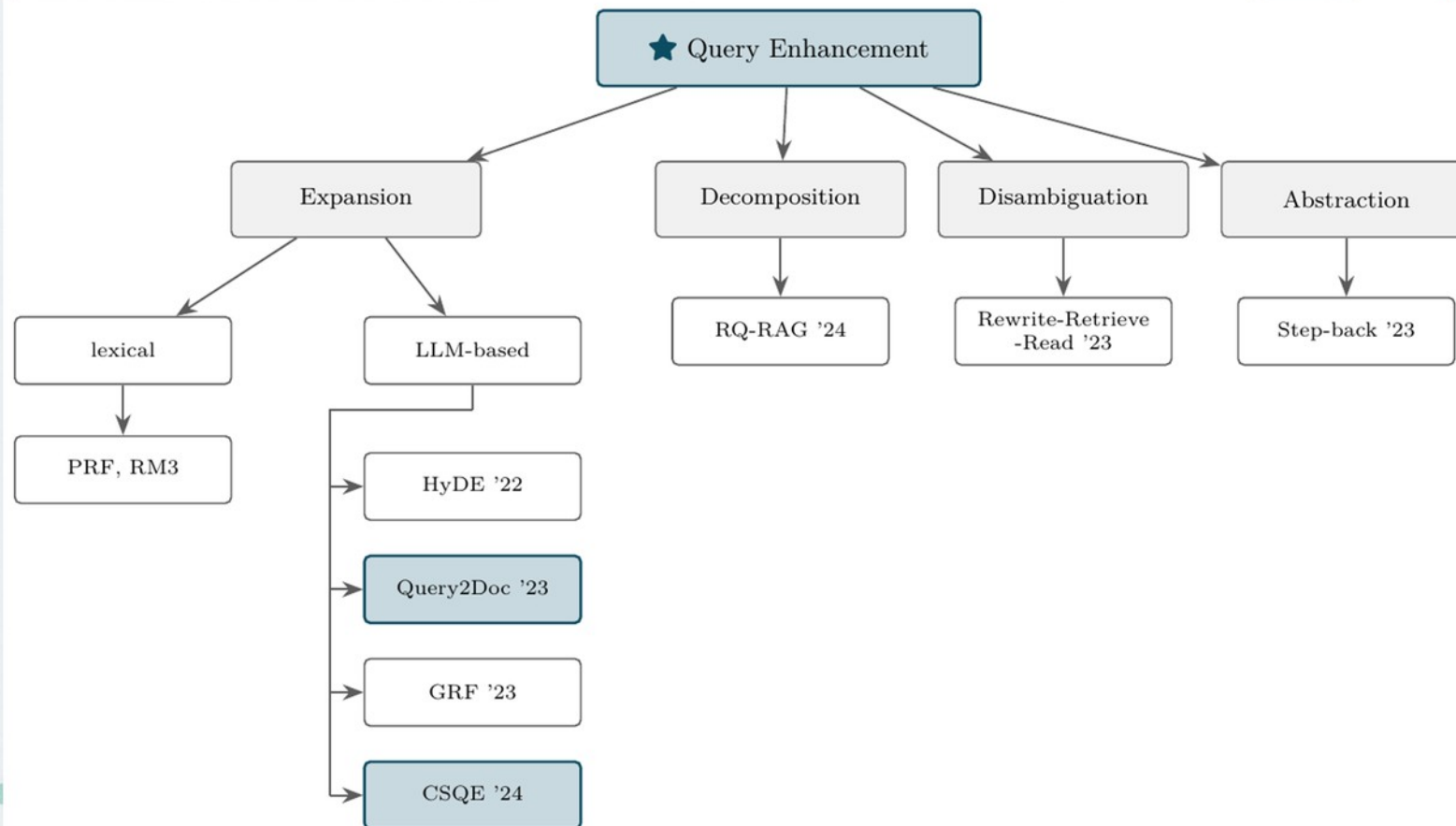
THE PROBLEM • ANSWER QUALITY CAN NEVER EXCEED RETRIEVAL QUALITY





THE GAP • RICH IN ENGLISH. UNVALIDATED IN ARABIC.

Query Enhancement — repair the query before it reaches the retriever



English

Dozens of published techniques



Arabic

Not one validated for Arabic

THE GAP • BUT ARABIC IS NOT ENGLISH



MORPHOLOGY

كتب • كاتب •
مكتوب • كتاب

One root, dozens of
surface forms



ORTHOGRAPHY

مكتبة / مكتبه
إسلام / اسلام

Same word, two
accepted spellings



DIACRITICS

كُتِبَ / كَتَبَ
عِلْمٌ / عَلَّمَ

Same letters,
different word



DIGLOSSIA

عايز أعرف شروط التأجيل

ما هي شروط تأجيل الدراسة

People ask in **dialect**,
knowledge is written
in **MSA**



RESEARCH QUESTION

“

To what extent can LLM-based query enhancement improve Arabic information retrieval, across sparse, dense and hybrid retrieval, using openly available models between 2 and 8 billion parameters?



Beyond validation – how do different retrievers interact with query enhancement?



OBJECTIVES

01

Establish dense and sparse baselines and measure their complementarity

02

Run a systematic error analysis to identify the failure modes

03

Adapt Query2Doc for Arabic zero-shot on free cloud GPUs

04

Compare ten open LLMs from 2 to 8 billion parameters

05

Analyse how expansions interact with sparse versus dense retrieval

06

Investigate query repetition as a remedy for term dilution

07

Establish a hybrid fusion baseline without query enhancement

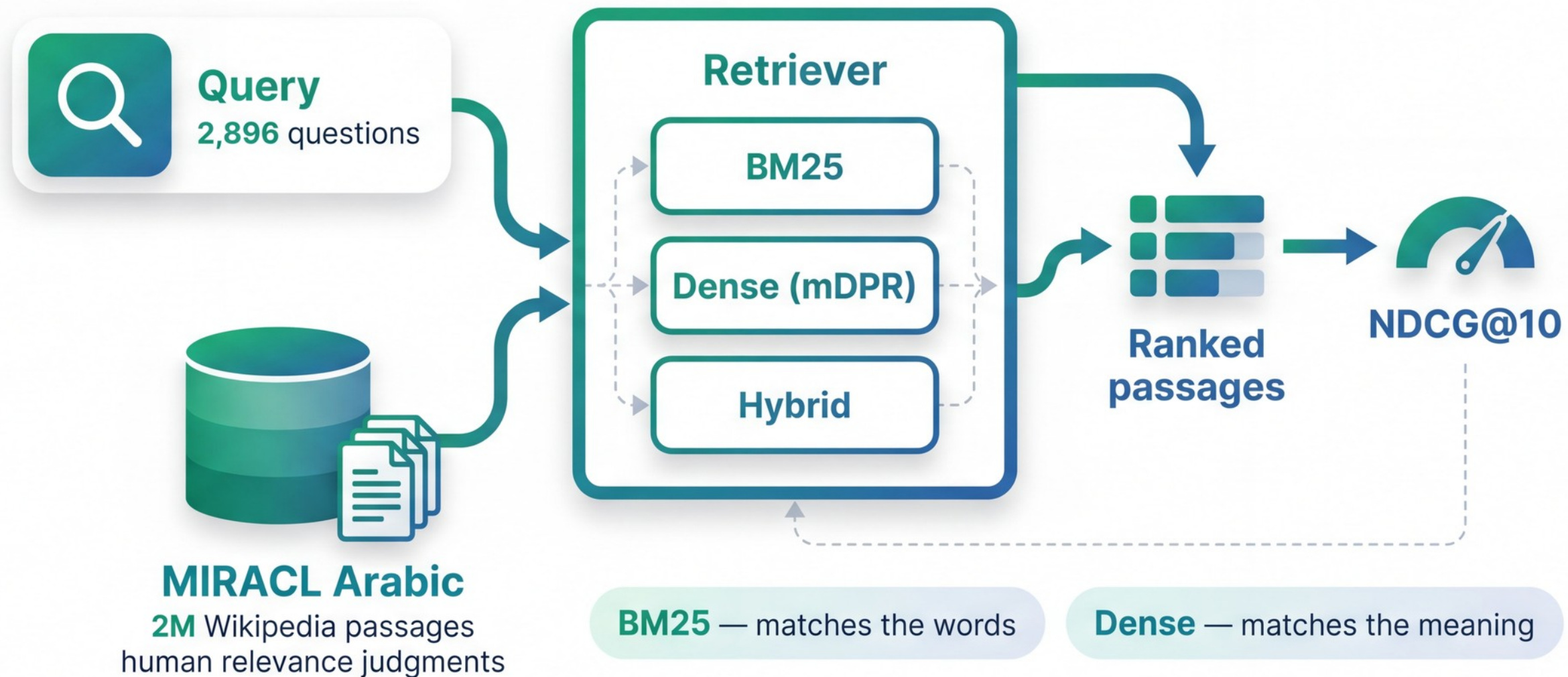
08

Adapt and evaluate CSQE for Arabic with component ablation

09

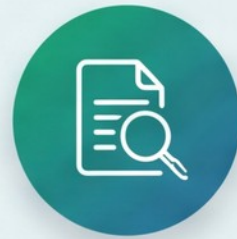
Determine the optimal placement of expansion in the hybrid pipeline

METHODOLOGY • THE SYSTEM WE BUILT





METHODOLOGY • BASELINE RESULTS, AND WHAT THEY POINTED AT



We **reproduced** the published MIRACL baselines before changing anything

0.4993

mDPR — dense baseline

0.4621

BM25S — sparse baseline

34%

of queries failed outright

0.345

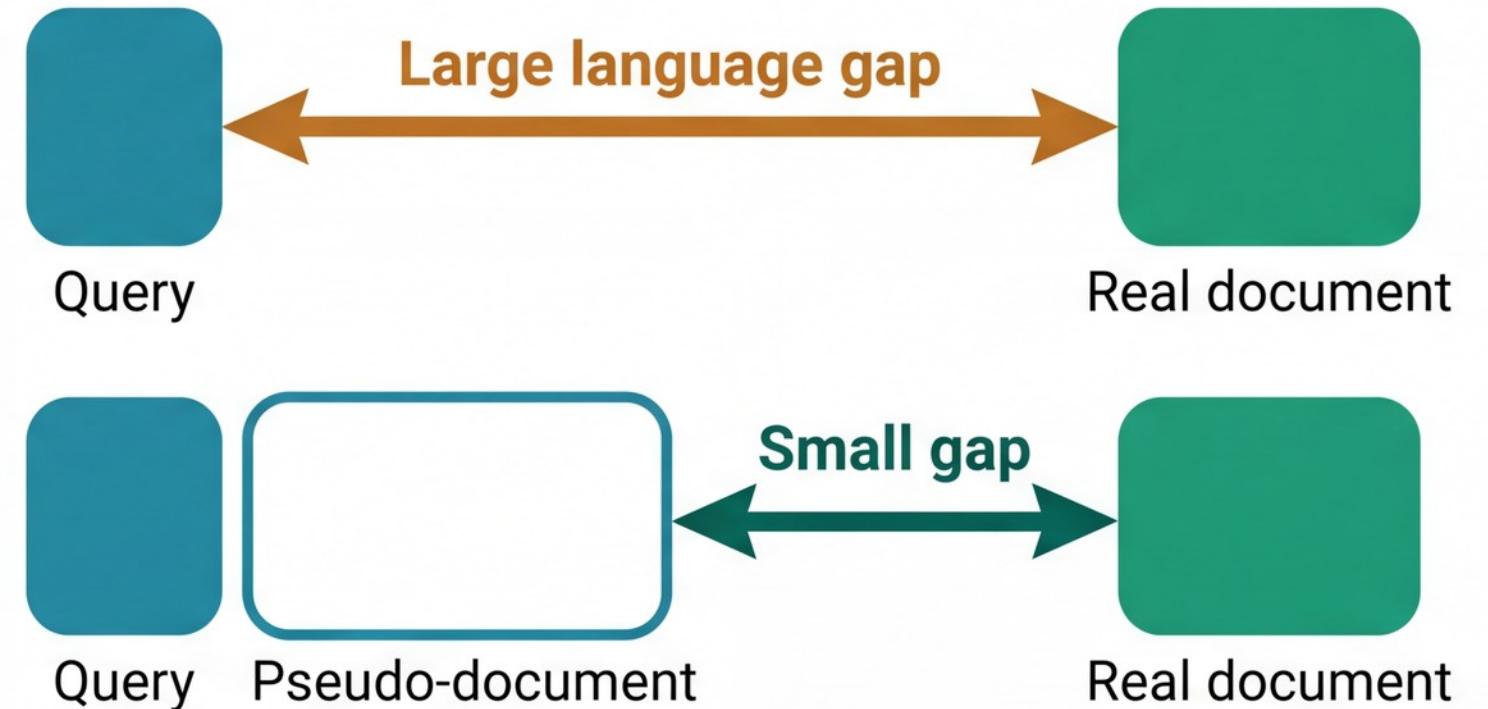
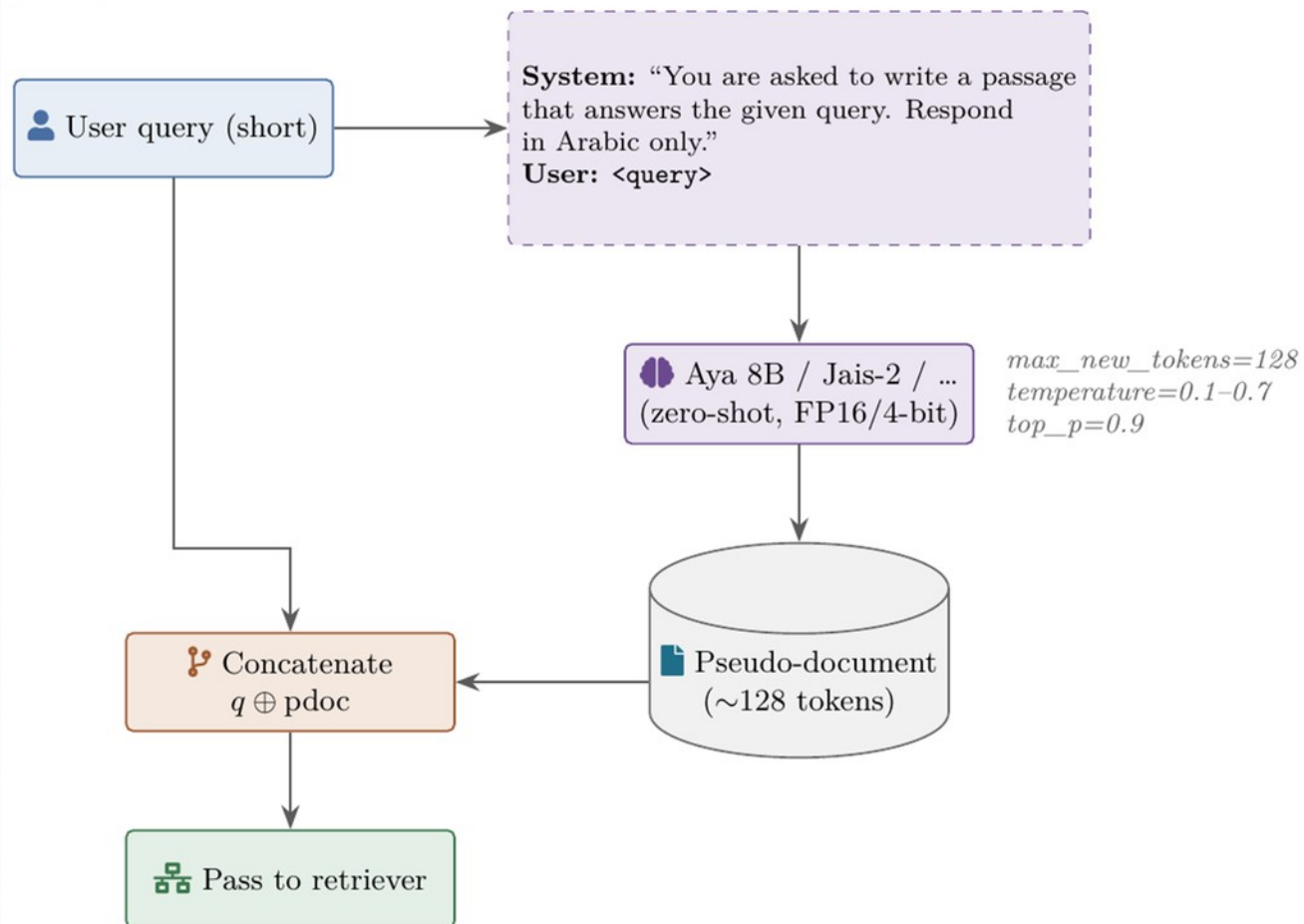
Short queries — the weakest group

Information poverty: too few words to separate one passage from two million



METHODOLOGY

QUERY2DOC - HALLUCINATION AS A FEATURE

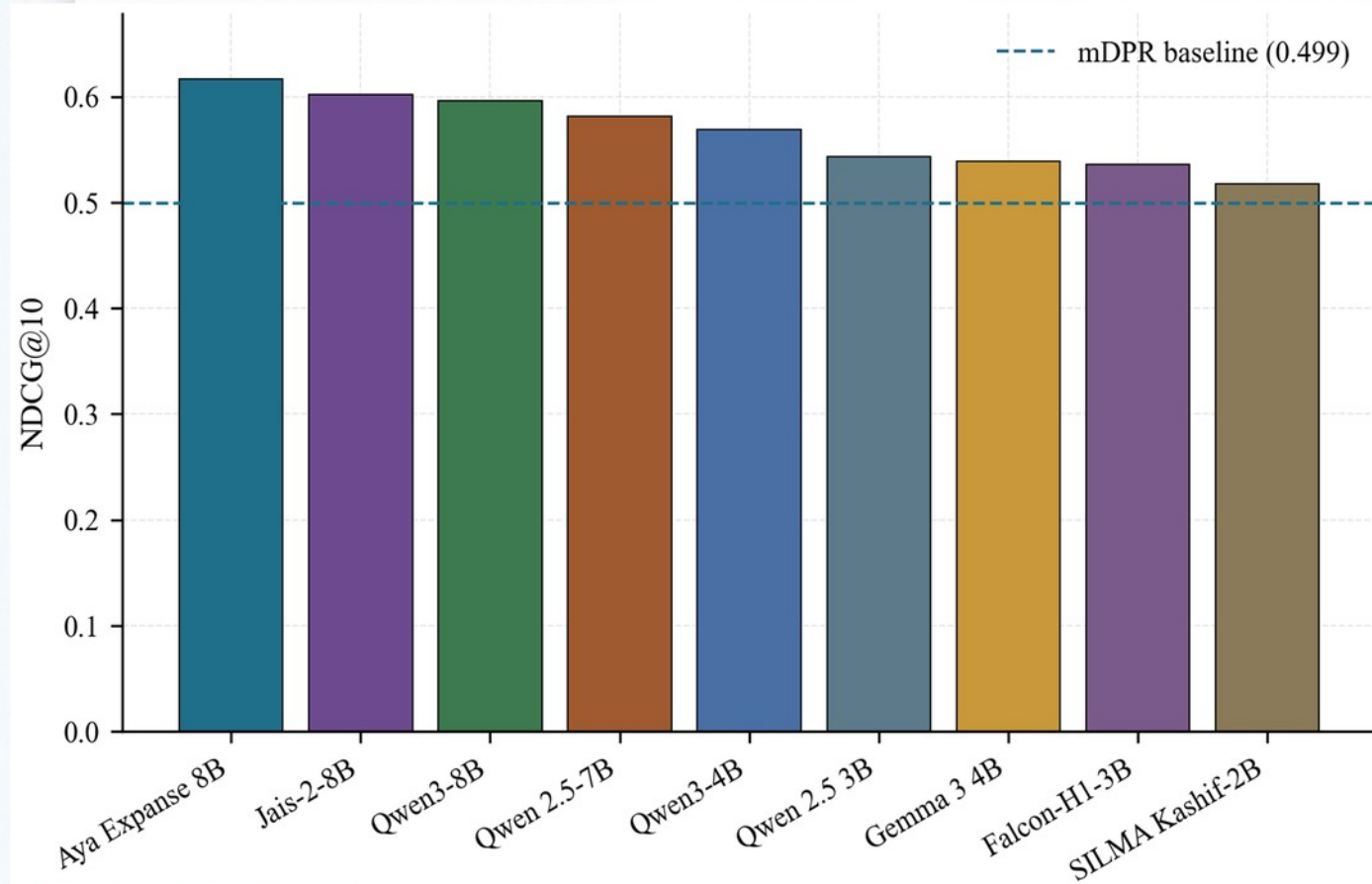


The guess may be **factually wrong**. Its **vocabulary** still matches the **corpus**.

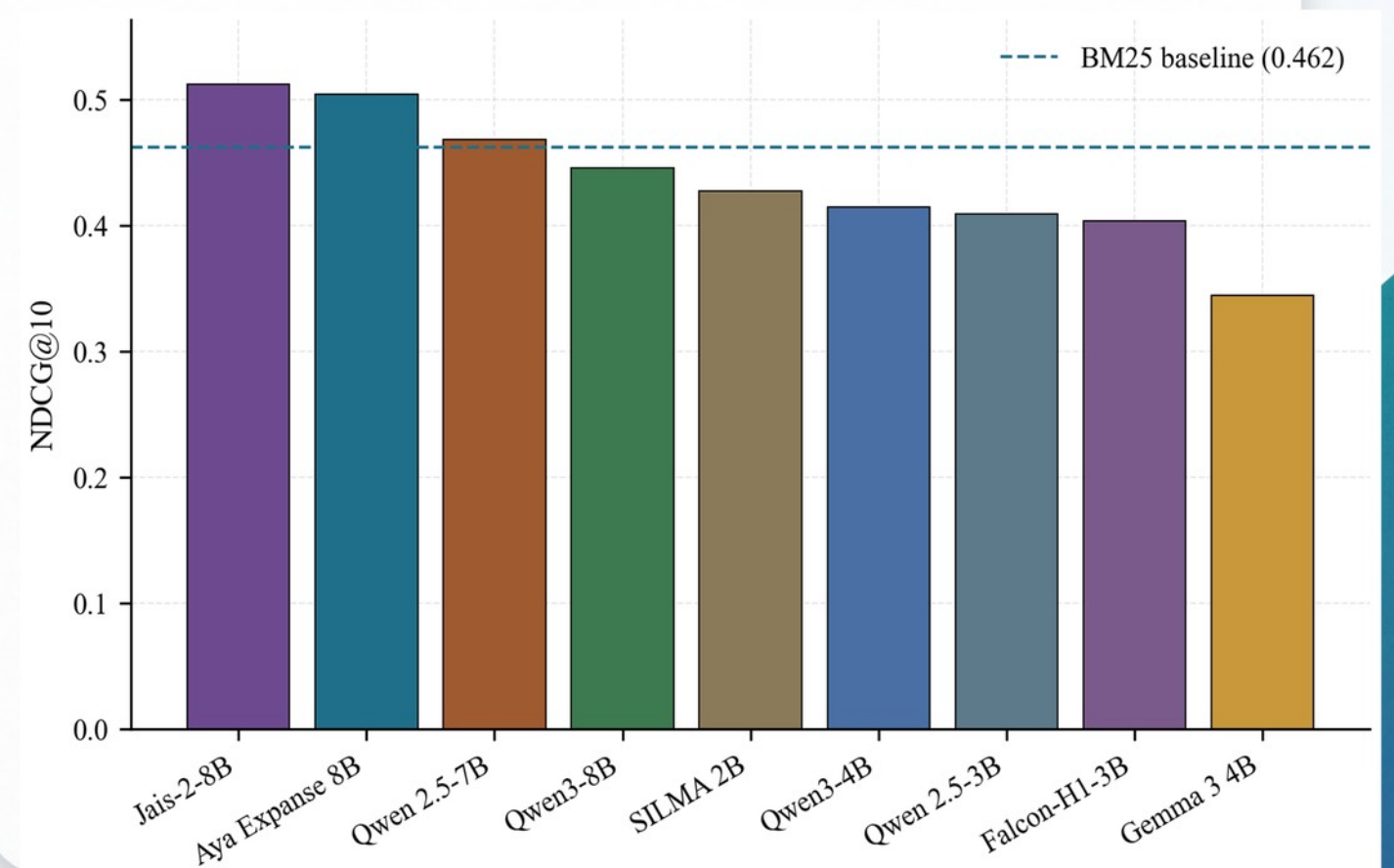


RESULTS • SAME EXPANSION. OPPOSITE EFFECT.

Dense — every model improved



BM25 — six of nine got worse



RESULTS • QUERY REPETITION RESTORES THE ORIGINAL QUERY WEIGHT



The original terms lose their weight

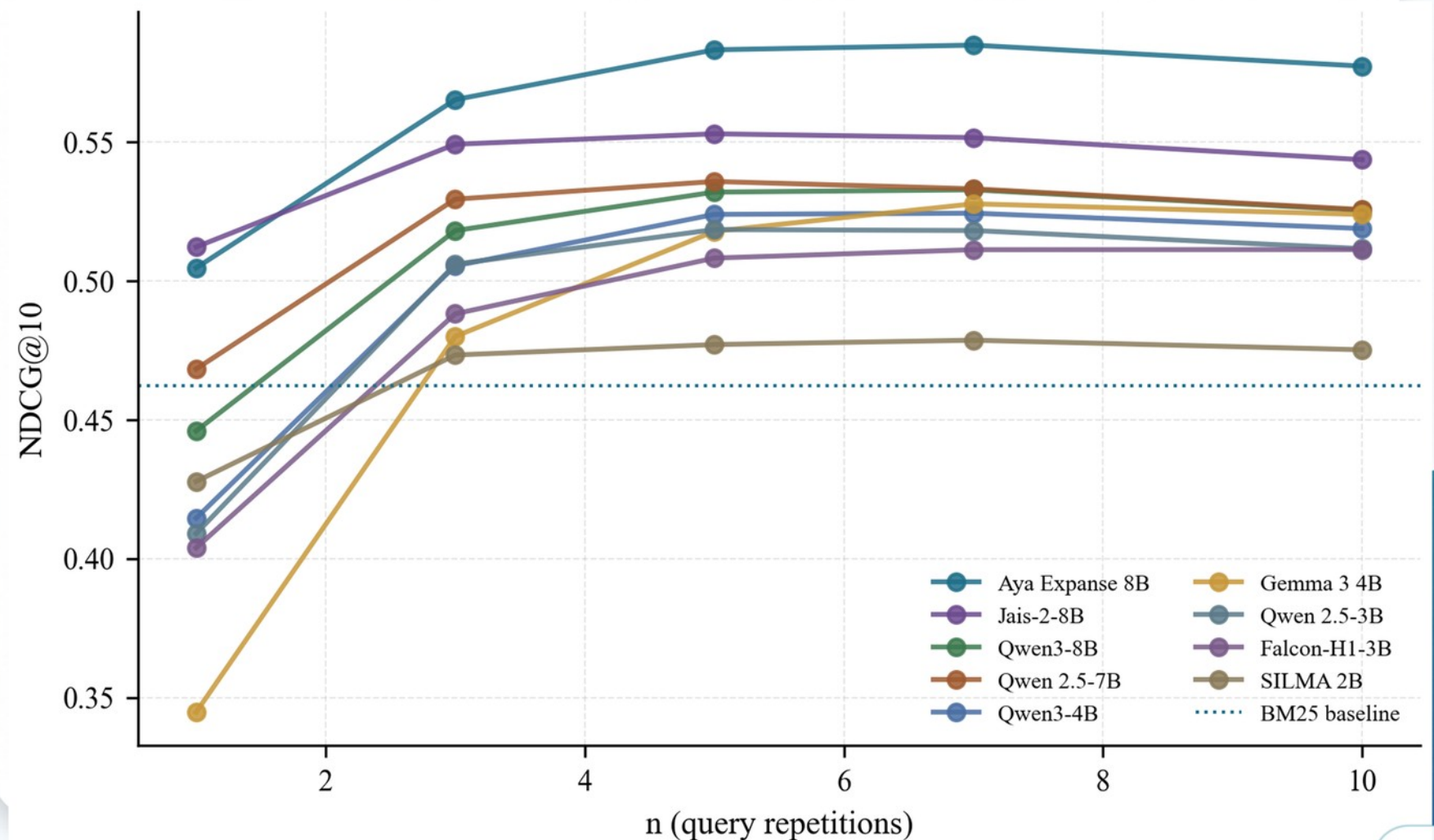


We repeat the query inside the expansion

0.5855

9 of 9 above baseline

Repetition sweep, nine models



THE CATCH • WHERE EXPANSION BREAKS



ما هي الأسماء الخمسة في اللغة العربية؟

A well-known rule of Arabic grammar



The correct answer

أب ، أخ ، حم ، فو ، ذو



What the model generated

محمد ، آدم ، إبراهيم ، إسماعيل

It listed people's names

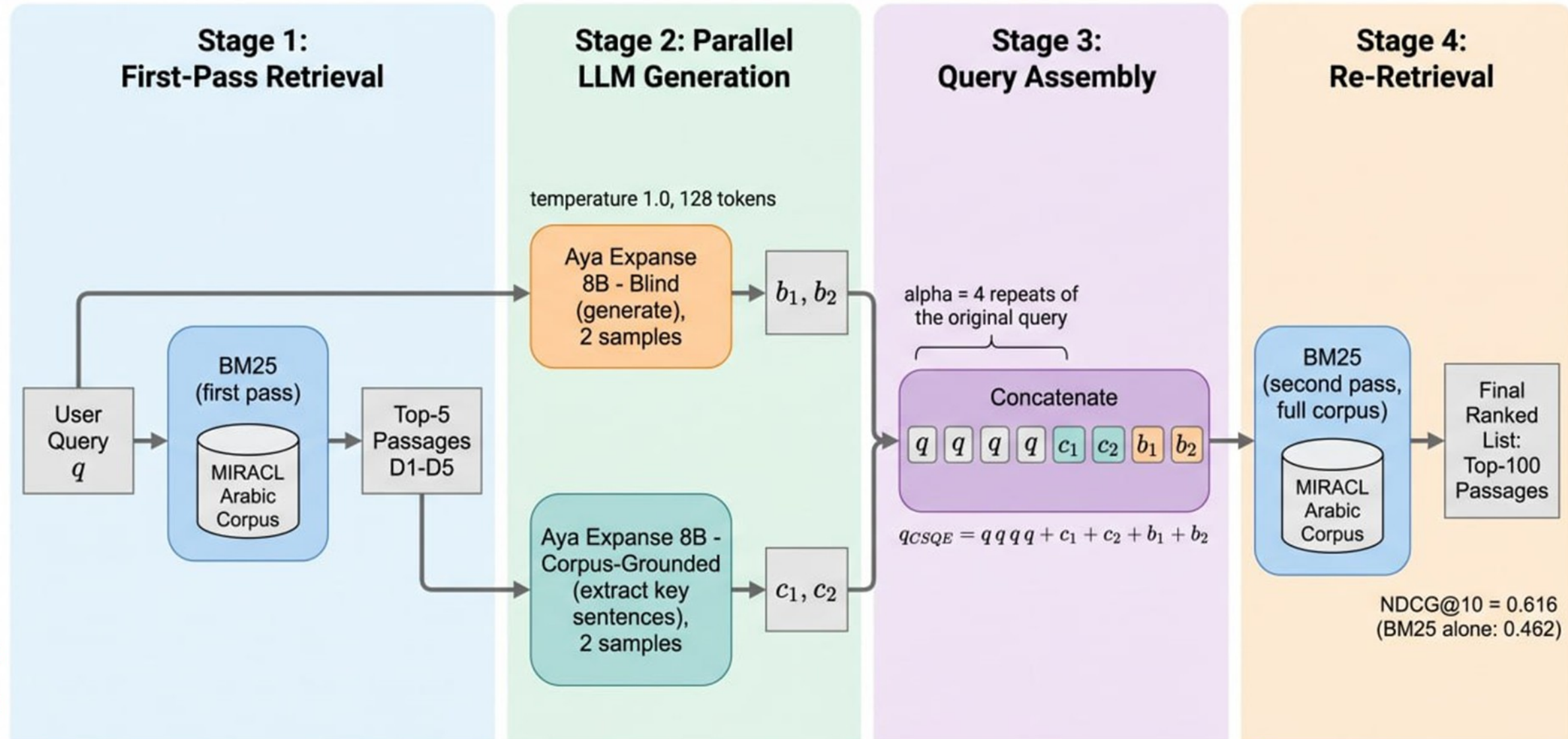
1.000
bare query

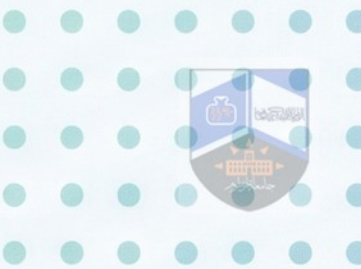


0.000
query + expansion



OUR METHOD • CORPUS-STEERED QUERY EXPANSION





THE CONTRIBUTION • WHICH RETRIEVER SHOULD GET THE EXPANDED QUERY?

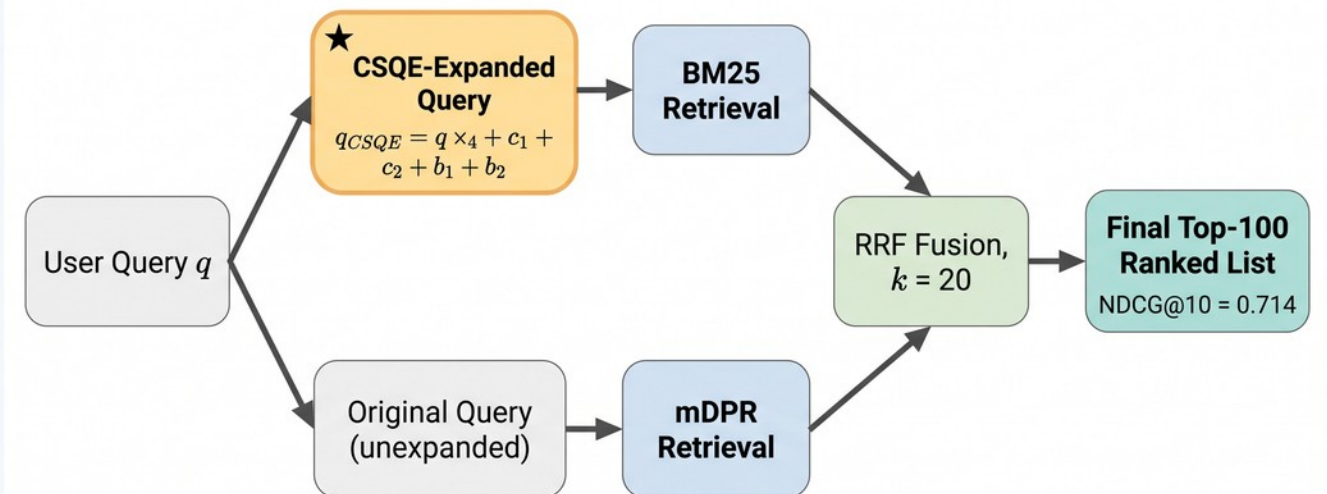
No expansion - hybrid only 0.6267

Dense retriever only 0.6474

Both retrievers 0.6936

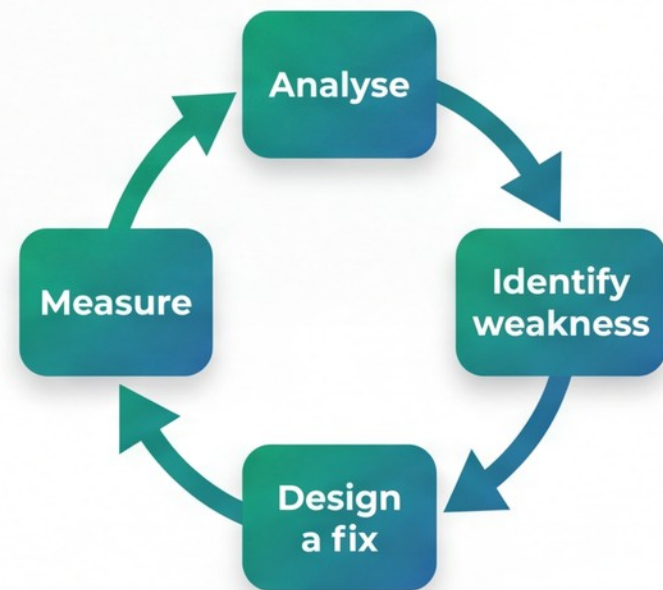
 BM25 only 0.7137

 **Fusion only pays** while the two retrievers **stay different**.
Expand both and they converge.





RESULTS • THE FULL JOURNEY



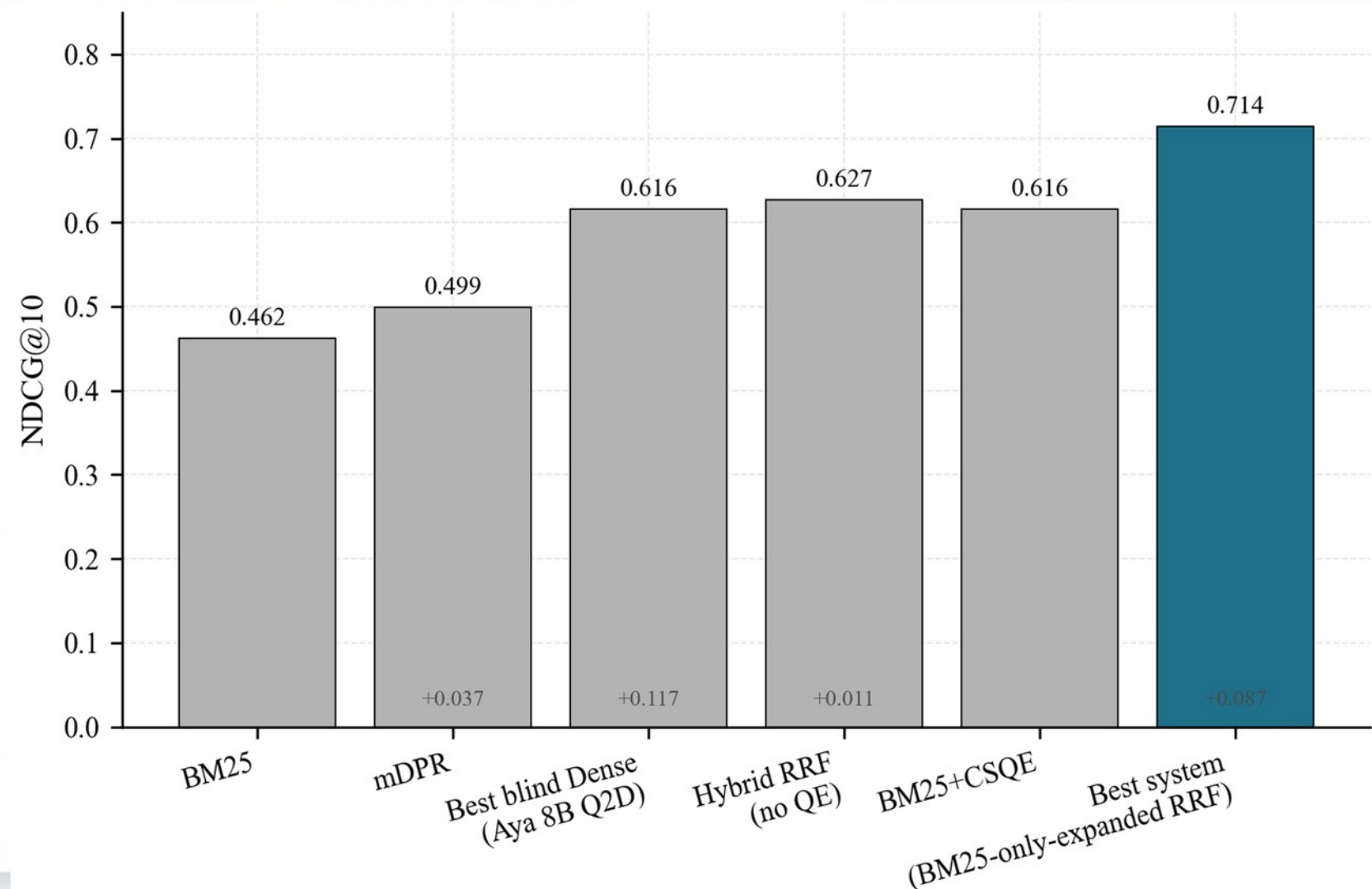
Our method, repeated

0.7137 final system

+54.5% over BM25

+13.9% over no-QE hybrid

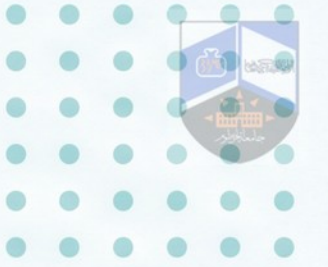
Baseline to final system





RESULTS • THE SAME TREND ACROSS ALL FOUR METRICS

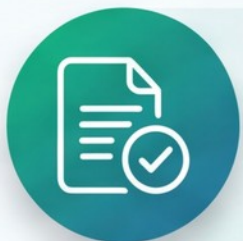
Method	NDCG@10	Recall@10	Recall@100	MRR
BM25S alone	0.4621	0.5964	0.8577	0.4836
mDPR alone	0.4993	0.6156	0.8407	0.5328
Blind QE → Dense	0.6164	0.7256	0.9001	0.6493
Blind QE + repetition → BM25	0.5855	0.7128	0.9300	0.6165
Hybrid RRF, no QE	0.6267	0.7597	0.9466	0.6517
CSQE → BM25	0.6157	0.7447	0.9422	0.6380
CSQE → Dense	0.5915	0.7073	0.8816	0.6225
★ CSQE on BM25 + Dense raw	0.7137	0.8363	0.9734	0.7362



CONCLUSIONS

Query enhancement transfers to Arabic — and the mechanisms we found are not Arabic-specific

- 01** Baselines established and error-analysed
- 02** Query2Doc transfers to Arabic zero-shot
- 03** Ten open LLMs compared under one protocol
- 04** Dense and sparse respond differently to expansion
- 05** Query repetition resolves sparse degradation
- 06** Corpus grounding validated
- 07** Retriever-specific query representation is critical



First systematic validation of LLM-based query enhancement for Arabic retrieval — ten open models, free hardware, fully reproducible. A paper is in preparation.

FUTURE WORK



DIALECTAL ARABIC

Evaluate query enhancement on **regional Arabic varieties** beyond **Modern Standard Arabic**.



FIRST-PASS QUALITY GATE

Fall back to **blind expansion** when the **first retrieval pass** is **poor**.



STRONGER EMBEDDING MODELS

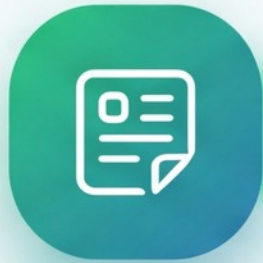
Test whether the gains hold over **more capable Arabic retrievers**.





Thank you

Questions



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